

Using the PytheM Framework *for* Web and Penetration Testing



With the increasing traffic in network based applications, security is becoming a challenge and the network environment has to be made safe from different types of attacks. The probability of vulnerabilities in network or Web based applications increases if vulnerability testing is not done properly. In traditional implementations, network administrators use their own sets of tools to test their network environment, but such tools can be restricted to specific types of attacks. It is always desirable that administrators use different types of penetration and vulnerability testing tools which are meant to resolve assorted attacks. This is done to check the overall deployment during different types of attacks, thus ensuring that the network or Web based environment is secured from multiple attacks without any compromise on safety.

Penetration testing and open source tools

A number of frameworks and software tools are available that provide the features to evaluate the network or Web based applications on different aspects and parameters. In a security audit of Web applications or network devices, the loopholes or vulnerabilities are checked from different aspects so that the attackers or sniffers cannot destroy the environment. Traditionally, penetration tools are used by the network administrators and application developers to analyse the weak points or vulnerabilities. In penetration testing, the applications or devices are put through pre-programmed

attacks so that the actual behaviour of hardware or software can be checked. If the applications or hardware devices react in abnormal ways during penetration testing, troubleshooting is done and suitable remedial measures are taken to cope with the attacks. Penetration testing can be implemented on any type of deployment including on networks, devices, websites, servers or software installations.

There are many penetration testing and security audit tools available that are free and open source, and can be used for security audits or bug tracking in the deployments. These tools provide the functions to analyse the applications or devices under different attacks and identify the weak points so that the vulnerabilities can be removed before actual deployment.

Python, a high performance programming language for multiple applications

Python is one of the powerful and cross-platform programming languages used for most of the high performance environments including cloud computing, Big Data processing, network programming, socket analytics, data science, statistical analysis and many others. It offers a large set of tools that have been developed specifically for penetration testing and security audits.

There are many penetration testing and security audit tools developed with Python, and these are used widely by corporates as well as individuals. These tools are used for digital forensics and vulnerability analysis on different types